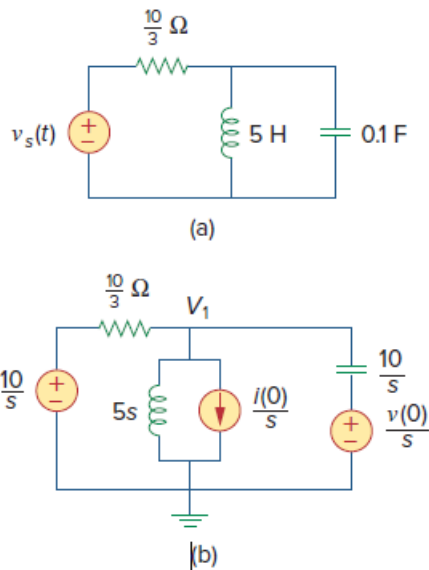


Problem

For the circuit shown in Fig. 16.12 with the same initial conditions, find the current through the inductor for all time $t > 0$.

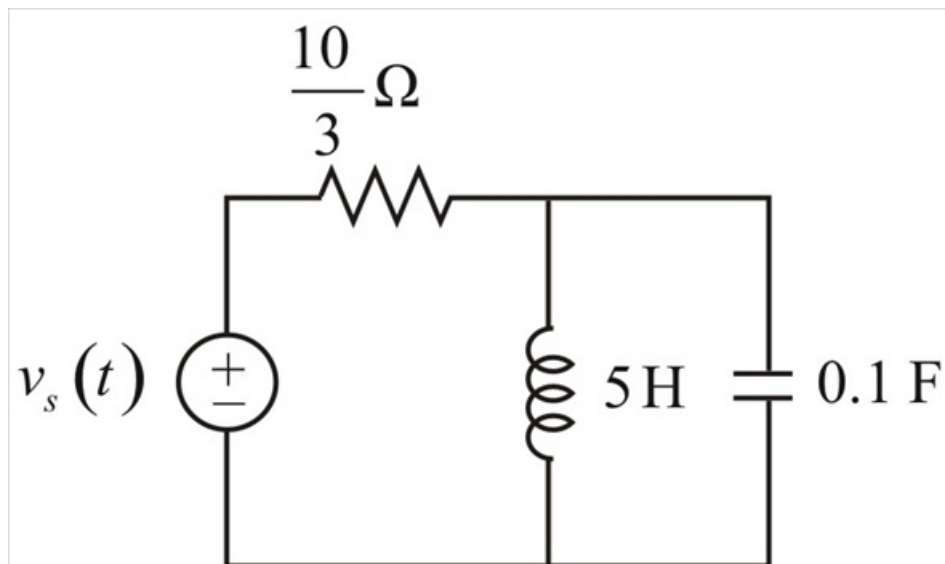
Figure 16.12:



Step-by-step solution

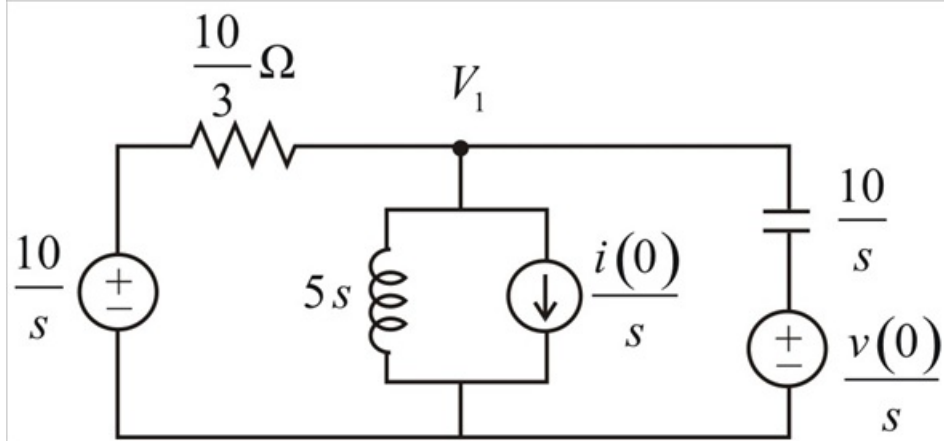
Step 1 of 6

Given circuit is



Step 2 of 6

The circuit in the s – domain with the initial conditions



Step 3 of 6

We know that V_1 is the voltage across the capacitor and also inductor in time domain.

$$\frac{V_1 - \frac{10}{s}}{\frac{10}{3}} + \frac{V_1}{5s} + \frac{i(0)}{s} + \frac{V_1 - \frac{v(0)}{s}}{\frac{10}{s}} = 0$$

$$V_1 \left(\frac{3}{10} + \frac{1}{5s} + \frac{s}{10} \right) = \frac{3}{s} - \frac{i(0)}{s} + \frac{v(0)}{10}$$

The initial conditions are $v(0) = 5 \text{ V}$ and $i(0) = -1 \text{ A}$

$$V_1 \left[\frac{3s + 2 + s^2}{10s} \right] = \frac{3}{s} + \frac{1}{s} + \frac{5}{10}$$

$$V_1 \left[\frac{s^2 + 3s + 2}{10s} \right] = \frac{30 + 10 + 5s}{10s}$$

$$V_1 = \frac{40 + 5s}{(s+2)(s+1)} \text{ ----- (a)}$$

Step 4 of 6

Using partial fraction expansion, we get

$$\frac{40+5s}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2} \text{ ---- (1)}$$

Multiplying $(s+1)(s+2)$ on both sides, becomes

$$40+5s = A(s+2) + B(s+1)$$

Consider $s = -1$

$$A = 40 - 5$$

$$A = 35$$

Consider $s = -2$

$$(-1)B = 40 - 10$$

$$B = -30$$

Substituting $A = 35$ and $B = -30$ in equation (1)

$$V_1 = \frac{35}{s+1} - \frac{30}{s+2}$$

Taking inverse transform

$$v_1(t) = (35e^{-t} - 30e^{-2t})u(t) \text{ V}$$

The current through the inductor for $t > 0$

Step 5 of 6

We know that

$$I_1 = \frac{V_1}{5s} + \frac{i(0^-)}{s}$$

Therefore,

$$I_1 = \left[\frac{1}{5s} \right] \left[\frac{40+5s}{(s+1)(s+2)} \right] - \frac{1}{s} \quad [\because \text{from (a)}]$$

$$I_1 = \frac{8+s}{s(s+1)(s+2)} - \frac{1}{s}$$

Applying partial fraction method,

$$\frac{8+s}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2} \text{ ---- (2)}$$

Multiplying $s(s+1)(s+2)$ on both sides we get

$$8+s = A(s+1)(s+2) + B(s+2)s + Cs(s+1)$$

Consider $s = 0$, then

$$(2)(1)A = 8$$

$$A = 4$$

Consider $s = -1$, then

$$(1)(-1)B = 8 - 1$$

$$B = -7$$

Consider $s = -2$, then

$$(-2)(-1)C = 8 - 2$$

Step 6 of 6

$$C = 3$$

Substituting values of $A = 4$, $B = -7$ and $C = 3$ in equation (2), we get

$$I_1 = \frac{4}{s} - \frac{7}{s+1} + \frac{3}{s+2} - \frac{1}{s}$$

Taking inverse transform, we obtain

At $t > 0$

$$i_1(t) = (3 - 7e^{-t} + 3e^{-2t})u(t) \quad \text{A}$$